Tender Submission — DesertVolt Industries

Medium-Voltage Metal-Enclosed Switchgear — 13.8 kV

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Executive Summary

DesertVolt Industries offers the DV-Shield14 indoor, metal-clad, withdrawable VCB switchgear platform. The equipment is designed for harsh, dusty environments typical of open-pit and plant-area substations in the Kingdom. The platform meets the Company Standard ES-EL-013.8KV-001 in environmental, electrical, safety, and documentation domains, with particular emphasis on internal arc containment and maintainability.

The proposal includes a complete secondary systems package based on numerical relays supporting IEC 61850 Edition 2, redundant Ethernet interfaces, and comprehensive FAT demonstrations. The design aligns with a room design temperature of +50 C and continuous operation at ambient up to +55 C.

[Ref §3] Service Conditions

The DV-Shield14 is qualified for −5 C to +55 C ambient, altitude up to 1000 m, and relative humidity up to 95 percent non-condensing. Dust ingress control is achieved using replaceable intake filters. Thermal management and ventilation calculations are provided for a +50 C switchroom.

[Ref §4] Electrical Ratings

The equipment is of the 17.5 kV class for 13.8 kV service. Insulation levels are LI (BIL) 95 kV and AC withstand 38 kV for 1 minute. Short-circuit rating is 31.5 kA for 1 s. The main busbar rating is 2500 A continuous with temperature rise within IEC limits in the specified environment.

[Ref §5] Internal Arc and Personnel Protection

DV-Shield14 has an internal arc classification of IAC AFLR at 31.5 kA for 1 s. The lineup incorporates top exhaust plenum connections suitable for ducting to a safe area. A concept routing is shown in the GA; final routing and termination locations are coordinated with the building design during detailed engineering.

[Ref §6] Ingress Protection and Compartmentation

External enclosure protection is IP4X with tested documentation available. Internal separation is IP2X between accessible compartments and live parts. Construction corresponds to a metal-clad arrangement equivalent to Form 3b.

[Ref §8] Secondary Systems and Protection

Protection relays (SIPROTEC/RELION families) support IEC 61850 Edition 2 with GOOSE and MMS. Redundant Ethernet interfaces are standard. Disturbance records are exported as COMTRADE files. Time synchronization supports SNTP and PTP.

[Ref §9] Earthing and Bonding

A full-length copper earth bar is provided. All doors, VT/CT cases, and withdrawable elements are bonded. Connection to the substation earth grid follows project drawings. A target earth resistance of 1 ohm or better is assumed.

[Ref §10] Installation Environment and Seismic

Clearances, access routes, and lifting points are included in the GA. Ventilation openings and heat rejection are sized for a +50 C room. Seismic anchorage is designed to the Saudi Building Code (SBC) Category S1; site-specific confirmation is provided once coordinates and soil data are finalized.

[Ref §11] Quality, Coatings, and Testing

An Inspection and Test Plan (ITP) defines FAT witness points for HV withstand, functional checks, wiring verification, and IEC 61850 demonstrations. Coatings are specified for ISO 12944 C4/C5 environments with documented surface preparation and dry film thickness measurements.

[Ref §12] Documentation Deliverables

Deliverables include GA drawings, SLDs, wiring schematics, protection and coordination study with CT saturation checks, arc-flash assessment and label set aligned with the site template, relay ICD/CID files, FAT/SAT protocols, O&M manuals, and a recommended spares list.

Appendices — Evidence Excerpts

Datasheet highlights: "+55 C", "IP4X", "2500 A".

GA notes: "Duct routing concept shown; final routing by site".

Document Register

- DRW-EL-SWGR-GA-1001 — Switchgear General Arrangement — Rev A — PDF
- MDS-EL-VCB-DV14 — VCB and Panel Datasheet — Rev B — PDF
- CAL-EL-PROT-COORD — Protection and Coordination Study — Rev 0 — PDF
- ITP-EL-SWGR-FAT — FAT/SAT Inspection and Test Plan — Rev 1 — PDF
- REP-EL-ARC-FLASH — Arc-Flash Hazard Assessment — Rev 0 — PDF